Name : Dharani K
Roll no:241901025

8.Collection of Email Evidence from a Victim's PC

Aim:

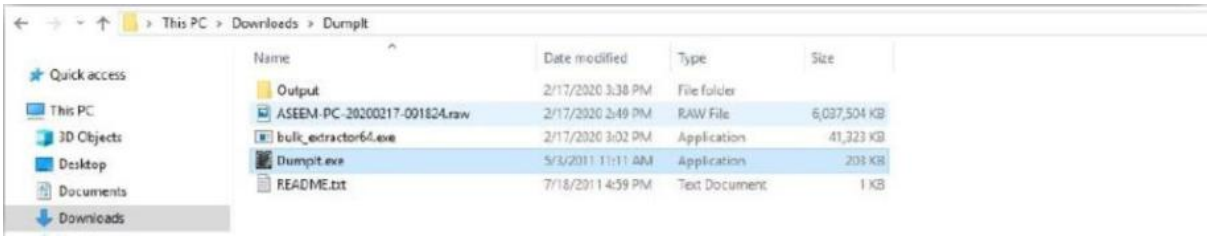
To study the methods used to identify, collect, preserve, and examine email evidence from a computer for digital forensic investigation.

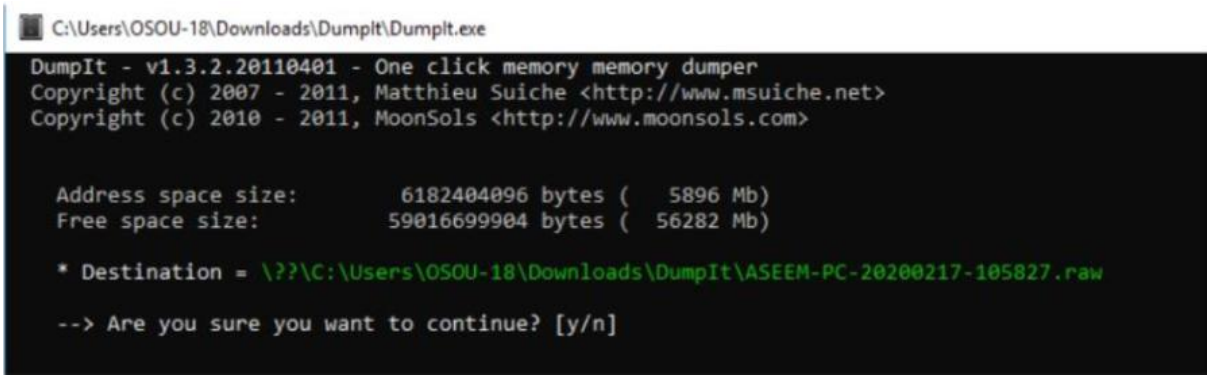
Procedure:

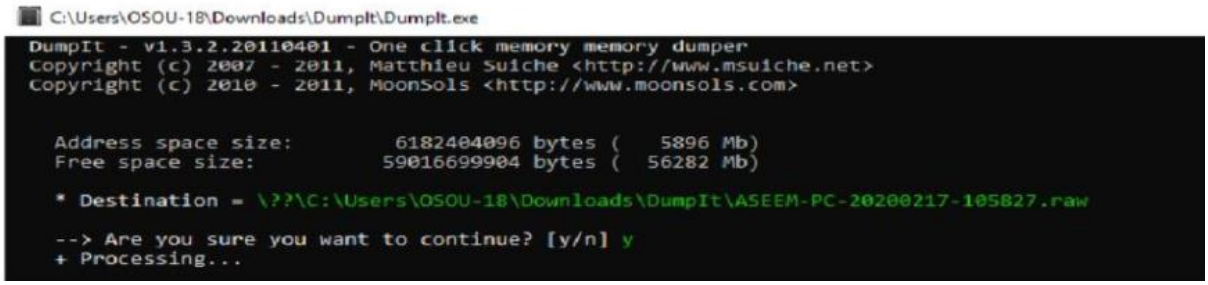
1. Start the computer and obtain proper authorization to examine the email evidence.
2. Identify the email service or client used on the system, such as Gmail, Microsoft Outlook, or Mozilla Thunderbird.
3. Record the system details, date, time, user account information, and other relevant case details.
4. If the email account is accessed through a web browser, examine the available email messages and relevant browser artifacts, following the approved evidence-handling procedure.
5. If a desktop email client is used, identify its local mail storage files. For example, Outlook may use .pst or .ost files, while Thunderbird may store messages in MBOX files or other profile data.
6. Create a forensic image of the storage device or an authorized copy of the relevant email data, using a suitable forensic tool.
7. Preserve the original evidence and perform the examination on a verified copy whenever possible.
8. Export relevant email messages in a suitable format, such as .eml or .msg, where supported.
9. Examine the email sender, recipient, subject, date, time, message body, and available message headers.

10. Review attachments carefully and record their file names, types, and hash values where appropriate.
11. Calculate cryptographic hash values, such as SHA-256, for collected evidence files and maintain an evidence log.
12. Document the collection method, tools used, timestamps, and findings in the forensic report.

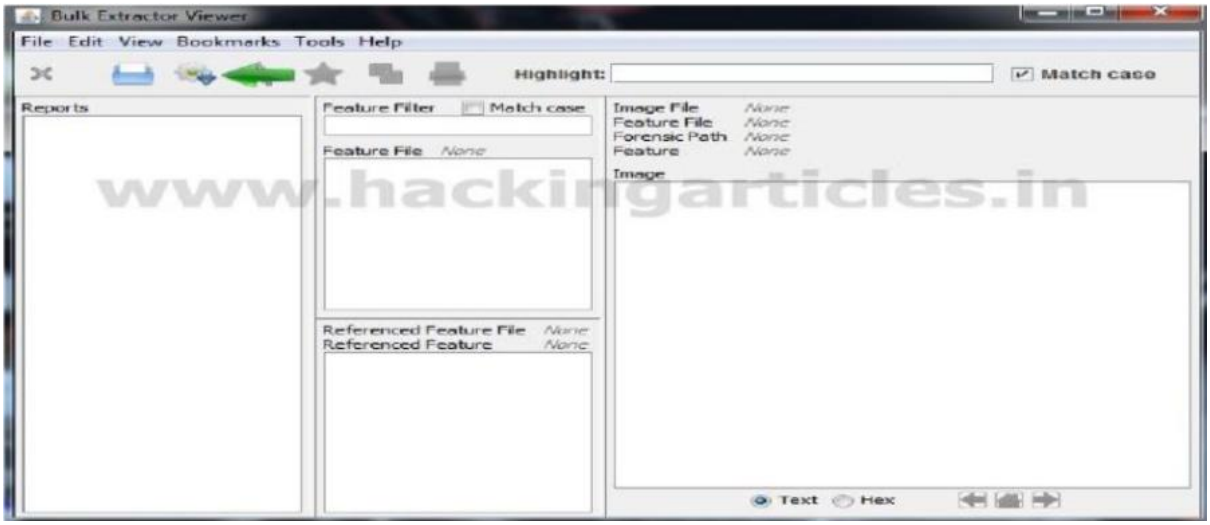
Output:



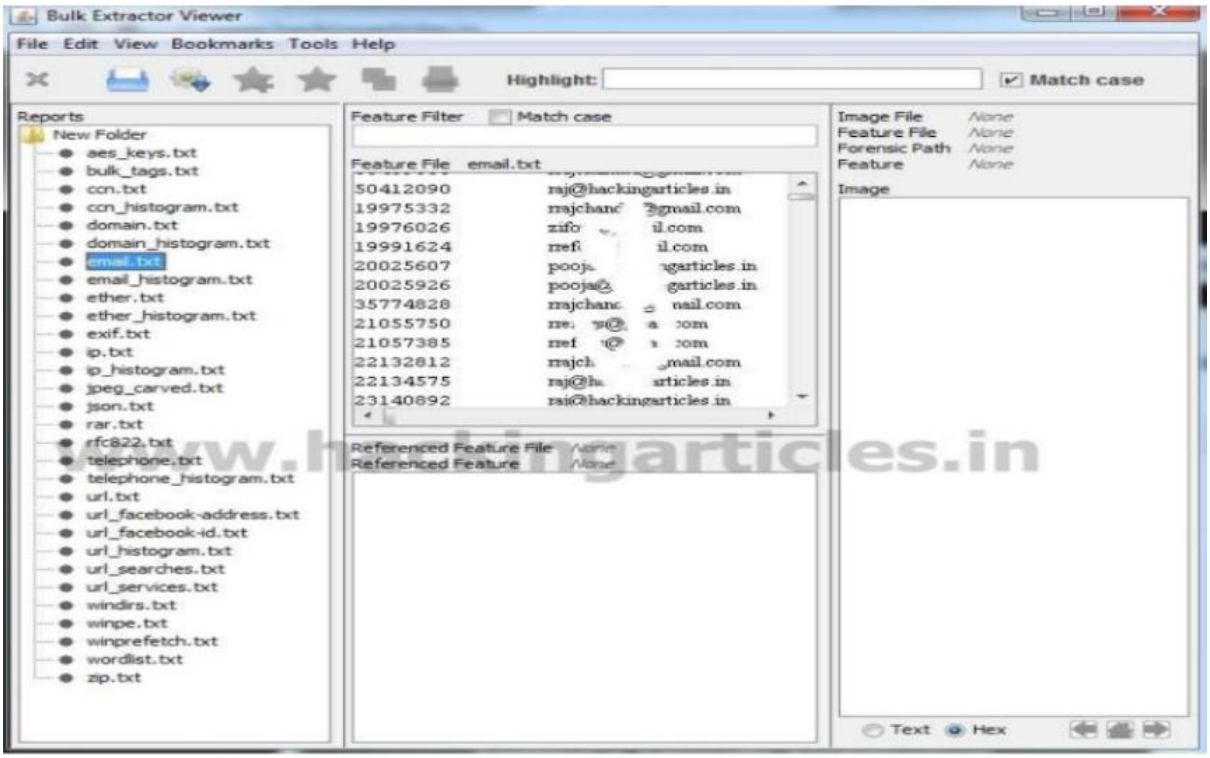


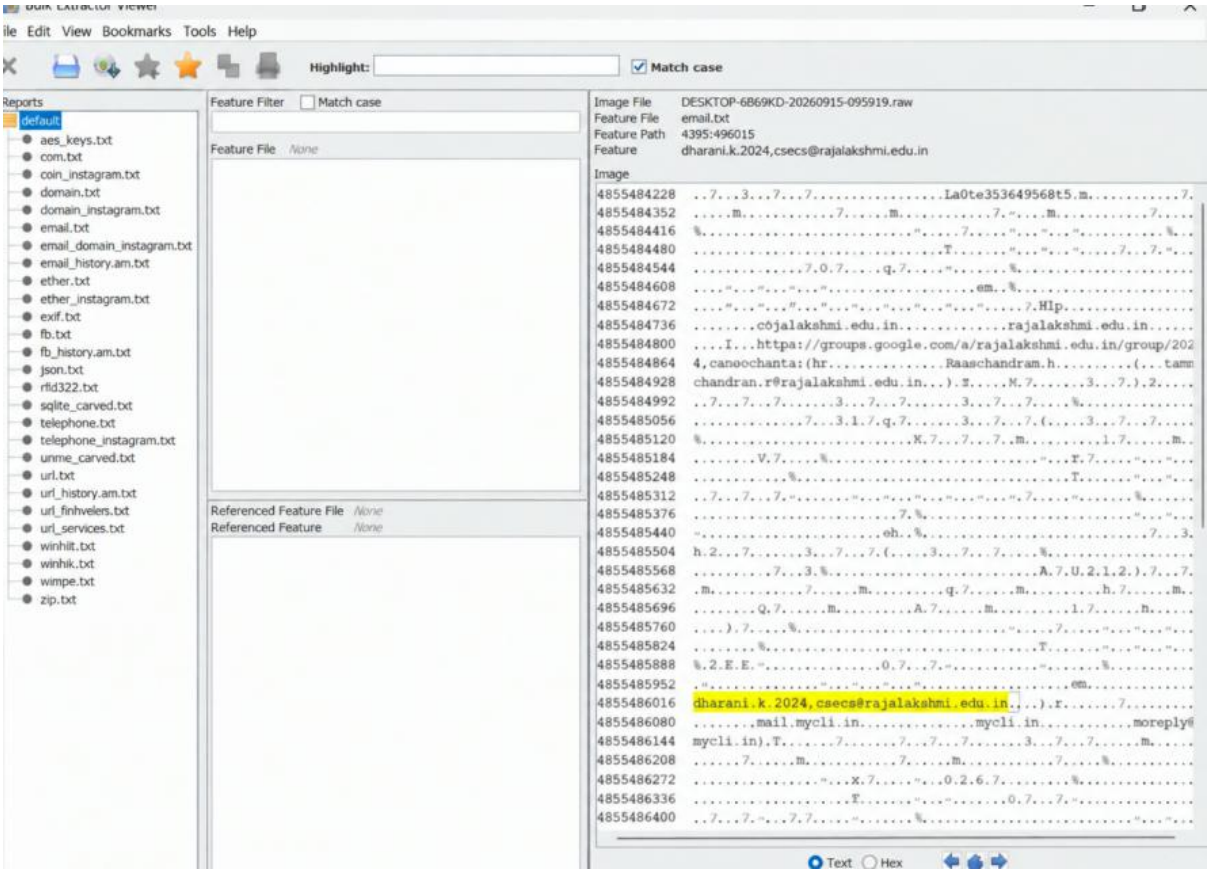


Name	Date modified	Type	Size
Output	2/17/2020 3:38 PM	File folder	
ASEEM-PC-20200217-105827.raw	2/17/2020 4:29 PM	RAW File	6,037,504 KB
bulk_extractor64.exe	2/17/2020 3:02 PM	Application	41,323 KB
DumpIt.exe	5/3/2011 11:11 AM	Application	203 KB
README.txt	7/18/2011 4:59 PM	Text Document	1 KB






Bulk Extractor Viewer
File Edit View Bookmarks Tools Help
Highlight:
Match case
Reports
New Folder
aes_keys.txt
bulk_tags.txt
ccn.txt
ccn_histogram.txt
domain.txt
domain_histogram.txt
email.txt
email_histogram.txt
ether.txt
ether_histogram.txt
exif.txt
ip.txt
ip_histogram.txt
jpeg_carved.txt
json.txt
rar.txt
rfc822.txt
telephone.txt
telephone_histogram.txt
url.txt
url_facebook-address.txt
url_facebook-id.txt
url_histogram.txt
url_searches.txt
url_services.txt
windirs.txt
winpe.txt
winprefetch.txt
wordlist.txt
zip.txt
Feature Filter
Match case
Feature File email.txt
50412090
19975332
19976026
19991624
20025607
20025926
35774828
21055750
21057385
22132812
22134575
23140892
Referenced Feature File
Referenced Feature
Image File
Feature File
Forensic Path
Feature
Image
Text
Hex


ile Edit View Bookmarks Tools Help
Highlight:
Match case
Reports
default
aes_keys.txt
com.txt
coin_instagram.txt
domain.txt
domain_instagram.txt
email.txt
email_domain_instagram.txt
email_history.am.txt
ether.txt
ether_instagram.txt
exif.txt
fb.txt
fb_history.am.txt
json.txt
rfid322.txt
sqlite_carved.txt
telephone.txt
telephone_instagram.txt
unme_carved.txt
url.txt
url_history.am.txt
url_finhvelers.txt
url_services.txt
winhilt.txt
winhik.txt
wimpe.txt
zip.txt
Feature Filter
Match case
Feature File None
Referenced Feature File None
Referenced Feature None
Image File DESKTOP-6B69KD-20260915-095919.raw
Feature File email.txt
Feature Path 4395:496015
Feature dharani.k.2024,csecs@rajalakshmi.edu.in
Image
4855484228
4855484352
4855484416
4855484480
4855484544
4855484608
4855484672
4855484736
4855484800
4855484864
4855484928
4855484992
4855485056
4855485120
4855485184
4855485248
4855485312
4855485376
4855485440
4855485504
4855485568
4855485632
4855485696
4855485760
4855485824
4855485888
4855485952
4855486016
4855486080
4855486144
4855486208
4855486272
4855486336
4855486400
dharani.k.2024,csecs@rajalakshmi.edu.in
Text
Hex

Observation:

Email evidence may contain important information about phishing, fraud, unauthorized access, harassment, and other cyber incidents. The evidence must be collected carefully to preserve its integrity.

S. No.	Evidence Type	Information Obtained
1	Sender and Recipient	Email addresses involved in communication
2	Subject	Purpose or topic of the email
3	Date and Time	Available message timestamps
4	Email Header	Routing details and technical metadata
5	Message Body	Content of the communication
6	Attachments	Files shared through the email
7	Local Mail Files	Stored email data from the email client
8	Hash Value	Used to verify whether collected files have changed
9	Evidence Log	Details of collection and handling

Email evidence may be available in a mail server, webmail account, local email client, browser artifacts, or backups. The exact information depends on the email service and the data retained by the system.

Result:

Thus, the methods for collecting and preserving email evidence from a computer were studied. The importance of examining email headers, message content, attachments, and metadata while maintaining evidence integrity was understood.